Reproduction of the Axton 1986 Thermal Neutron Constants Evaluation

Introduction

This report documents the reproduction of the iterative Generalized Least Squares (GLS) evaluation of thermal neutron constants performed by E.J. Axton in 1986 (report GE/PH/01/86, CBNM, JRC Geel). The original evaluation determined best estimates and uncertainties for the thermal neutron cross sections, neutron yields, and related quantities of the fissile nuclides 233U, 235U, 239Pu, 241Pu, and the fission neutron yield of 252Cf.

Two independent data sources are available for this reproduction:

1. **JSON database** (`axton1986_data.json`): 167 measurements manually transcribed from the original Axton 1986 PDF report, with named correlation codes, metadata (author, reference, measured function), and systematic uncertainty decomposition.
2. **AGS/thermalcst files** (`ags_axton1986_jan2026.mic`, `thermalcst_axton1986_jan2026.mic`): Data files provided by a colleague at CEA Cadarache, containing 102 microscopic measurements. The thermalcst files hold measured values and absolute uncertainties; the AGS files hold the uncertainty decomposition for covariance matrix construction.

The report proceeds in six sections, each building on the previous. We begin by establishing agreement with Axton's published all-data fit, then progressively narrow the measurement set to understand how the microscopic subset was defined, and finally cross-validate the two data pipelines against each other.

Quantities and Notation

The evaluation fits the following primary quantities for each nuclide (identified by mass number: 33=233U, 34=234U, 35=235U, 39=239Pu, 41=241Pu, 52=252Cf):

Symbol	Description
SCA	Scattering cross section (coherent + incoherent)
SCR	Scattering cross section (coherent, "real")
ABS	Absorption cross section
FIS	Fission cross section
NUB	Prompt + delayed fission neutron yield (nu-bar)
WGA	Westcott g-factor for absorption
WGF	Westcott g-factor for fission
CAP	Capture cross section (= ABS - FIS for fissile nuclides)
CA	Capture cross section (fitted directly for non-fissile 240Pu, 242Pu)
HLF	Half-life (fixed, not fitted)
GC116	Westcott g-factor correction at 116 eV (absorption)
GA116	Westcott g-factor correction at 116 eV (fission)

Derived quantities computed from fitted parameters:

Symbol	Formula
CA(n)	ABS(n) - FIS(n)
ETA(n)	NUB(n) * FIS(n) / ABS(n)
ALPHA(n)	[ABS(n) - FIS(n)] / FIS(n)
F3ETA(n)	NUB(n) * FIS(n) * WGF(n) - ABS(n) * WGA(n)

Correlation Codes

The covariance matrix is constructed from systematic uncertainty components shared between measurements. Each component is identified by a correlation code (from Table 2 of the Axton 1986 report). Two measurements sharing a nonzero component for the same code are correlated; the relative covariance is cov_rel(i,j) = sum_k u_ik * u_jk, where u_ik is the relative systematic uncertainty (in percent) of measurement i for correlation code k.

Code	Description
`A`	Deruytter Common Uncertainty
`B`	Deruytter Timing
`C`	Deruytter-Berceano Assay
`D`	White U235 Assay
`E`	Maslin-Saplakoglu Lem./Deruyt. Correction
`F`	NPL Common Mn-bath Uncertainty
`G`	Uncertainty in (1+sigmaS/sigmaMn) in Mn-baths
`H`	Energy Uncertainty of U233 Neutrons
`I`	Energy Uncertainty of U235 Neutrons
`J`	Energy Uncertainty of PU239 Neutrons
`K`	Energy Uncertainty of Pu241 Neutrons
`L`	Energy Uncertainty of CF252 Neutrons
`M`	Detector slope (Different Authors)
`N`	Boldeman Detector Slope
`O`	Gwin Detector Slope
`P`	Hopkins Detector Slope
`Q`	Asplund Detector Slope
`R`	Colvin Detector Slope
`S`	Mather Detector Slope
`U`	U233 Delayed Neutrons
`V`	U235 Delayed Neutrons
`W`	PU239 Delayed Neutrons
`X`	PU241 Delayed Neutrons
`Y`	CF252 Delayed Neutrons
`Z`	Delayed Gamma-rays
`Delta`	Boldeman's Foil Thickness corrections
`A_`	Gwin Common Uncertainty
`B_`	Spiegel Bozorgmanesh Common Uncertainty
`C_`	Boldeman Common Uncertainty
`D_`	Diven Common Uncertainty
`E_`	Boron Pile Common Uncertainty

Code	Description
F_	Smith/Aleksandrov Common Uncertainty
G_	Smith Eta/Nubar Common Uncertainty
H_	Eta Montecarlo Corrections
I_	Mather Common Uncertainty
J_	White Pu239 Assay
K_	White PU241 Assay
M_	Cabell Fluence
N_	Cabell Temperature
O_	DIDO Reflector g+rs For CAP, ABS PU239
P_	DIDO Reflector g+rs For CAP, ABS PU241
Q_	CABELL 86 deg Spectrum Uncertainty
R_	Cabell Uncertainty (Sjostrand and Story)
S_	Cabell Uncertainty (Sjostrand and Story)
T_	Gwin Uncertainty evaluated By Hardy
U_	Muelhaus uncertainty evaluated by Axton
V_	Spectral Uncertainties from IN1060
W_	MTR 70 deg Spectrum Uncertainty
X_	NRU 40 deg Temperature Uncertainty
Y_	NRU Westcott r Uncertainty
Z_	Lounsbury Temperature Uncertainty
Delta_	Popovic Uncertainty (Na Cross Section)
rho	Keith Uncertainty (Deruytter correction)
dag	Uncertainty in Co Cross Section
dn	+/-20 deg Uncertainty in T
plus	Common Uncertainty in U235 Reactivity
sq	Common Uncertainty in U235 Reactivity
c	Half-life U233
hc	Half-life U234
n	Half-life PU239
u	Half-life AM241
alpha	Spencer Common Uncertainty

Note: There is no code `T` in the original Axton system (the sequence skips from `S` to `U`). Similarly, there is no code `L_`. These gaps are reflected in the AGS column numbering, where column 19 is unused.

---

1. All-Data Fit: Reproducing Axton 1986 Table 4

The first benchmark uses all 167 measurements from the JSON database with 38 free parameters, reproducing the conditions of Table 4 in the Axton 1986 report. This establishes the baseline: if the transcribed data is correct and the GLS implementation is faithful, the results should match Table 4 closely.

Script: `benchmark_axton1986_all_data.py`

Summary

Metric	This work	Report
Chi-squared	102.3	103
N measurements	167	167
N free parameters	38	38
Degrees of freedom	129	129

Fitted Parameters

Parameter	Fitted	Report	Diff%	Unc%	Rep Unc%	Match
SCA 33	12.1810	12.1861	-0.0419%	5.482	5.483	OK
SCA 35	15.9730	15.9828	-0.0615%	6.914	6.914	OK
SCA 39	7.9037	7.8966	+0.0894%	12.330	12.333	OK
SCA 41	12.1685	12.1878	-0.1586%	21.471	21.470	OK
SCR 33	10.9067	10.9169	-0.0934%	8.069	8.050	OK
SCR 35	14.1940	14.2071	-0.0919%	8.927	8.919	OK
SCR 39	6.8633	6.8000	+0.9308%	24.295	25.726	DIFF
SCR 41	11.1428	11.1717	-0.2584%	32.544	32.337	OK
ABS 33	576.2605	576.2174	+0.0075%	0.225	0.226	OK
ABS 35	681.8566	681.8303	+0.0039%	0.192	0.192	OK
ABS 39	1018.7581	1018.9667	-0.0205%	0.281	0.282	OK
ABS 41	1373.9632	1373.2025	+0.0554%	0.661	0.662	OK
FIS 33	530.7214	530.6953	+0.0049%	0.253	0.253	OK
FIS 35	582.8180	582.7836	+0.0059%	0.196	0.196	OK
FIS 39	747.6949	747.6236	+0.0095%	0.269	0.269	OK
FIS 41	1012.2387	1011.8730	+0.0361%	0.651	0.652	OK
NUB 33	2.4950	2.4950	-0.0006%	0.159	0.161	OK
NUB 35	2.4333	2.4334	-0.0042%	0.146	0.147	OK
NUB 39	2.8819	2.8822	-0.0094%	0.176	0.177	OK
NUB 41	2.9458	2.9463	-0.0155%	0.195	0.196	OK
NUB 52	3.7679	3.7676	+0.0091%	0.126	0.126	OK
WGA 33	0.9994	0.9995	-0.0084%	0.107	0.107	OK
WGA 35	0.9789	0.9789	+0.0030%	0.084	0.084	OK
WGA 39	1.0783	1.0782	+0.0049%	0.227	0.227	OK
WGA 41	1.0442	1.0442	+0.0026%	0.190	0.190	OK
WGF 33	0.9955	0.9955	+0.0009%	0.142	0.142	OK
WGF 35	0.9774	0.9774	-0.0027%	0.083	0.083	OK
WGF 39	1.0554	1.0555	-0.0063%	0.212	0.212	OK
WGF 41	1.0444	1.0445	-0.0113%	0.526	0.527	OK
CA 40	289.3345	289.3296	+0.0017%	0.482	0.482	OK
CA 42	18.5148	18.5144	+0.0020%	2.161	2.161	OK
CAP 34	95.8364	95.8369	-0.0005%	2.071	2.071	OK
GC116 39	1.3658	1.3265	+2.9593%	2.738	6.056	DIFF
GC116 40	1.0755	1.0860	-0.9671%	1.819	2.000	DIFF
GC116 41	1.0902	1.1085	-1.6477%	1.651	2.457	DIFF

Parameter	Fitted	Report	Diff%	Unc%	Rep Unc%	Match
GC116 42	1.1349	1.1335	+0.1219%	2.668	2.670	OK
GA116 39	1.1878	1.1846	+0.2726%	1.054	1.521	OK
GA116 41	1.1097	1.1073	+0.2159%	0.738	0.819	DIFF
CA 33	45.5391	45.5221	+0.0374%	1.537	1.535	OK
CA 35	99.0386	99.0467	-0.0082%	0.750	0.750	OK
CA 39	271.0633	271.3431	-0.1031%	0.787	0.786	DIFF
CA 41	361.7244	361.3294	+0.1093%	1.385	1.367	OK
ETA 33	2.2978	2.2979	-0.0036%	0.177	0.180	OK
ETA 35	2.0799	2.0799	-0.0017%	0.167	0.167	OK
ETA 39	2.1151	2.1147	+0.0202%	0.241	0.243	OK
ETA 41	2.1703	2.1710	-0.0328%	0.352	0.353	OK
ALPHA 33	0.0858	0.0858	+0.0070%	1.636	1.634	OK
ALPHA 35	0.1699	0.1700	-0.0408%	0.791	0.792	OK
ALPHA 39	0.3625	0.3629	-0.1014%	0.843	0.841	DIFF
ALPHA 41	0.3574	0.3571	+0.0703%	1.400	1.376	OK
F3ETA 33	742.2714	742.2525	+0.0025%	0.317	0.319	OK
F3ETA 35	718.5917	718.5735	+0.0025%	0.308	0.309	OK
F3ETA 39	1175.7724	1175.7377	+0.0030%	0.473	0.475	OK
F3ETA 41	1679.5072	1679.8787	-0.0221%	0.844	0.846	OK

Result: 47/54 parameters match within tolerance (0.1% for well-constrained parameters, 0.5% for poorly constrained). Chi-squared is 102.3 vs the reported 103. The small discrepancies in GC116 and GA116 parameters are attributable to rounding in the PDF-transcribed values.

The excellent agreement confirms that both the data transcription and the GLS implementation are faithful to the original evaluation.

---

2. Microscopic Fit: Narrow Definition

Having established the all-data baseline, we now attempt to reproduce the microscopic-only fit reported in Table 8 of the Axton 1986 report. Table 8 reports results for 101 measurements with 38 parameters, yielding chi-squared = 48.

The challenge is that the report does not explicitly list which measurements constitute the "microscopic" subset. This benchmark uses a narrow definition: exclude only the clearly Maxwellian-derived functions (FA, FF, CAP, FFH, F1ETA, F2ETA, F3ETA, FLEM, F1BIG, F1CAB-F5CAB) while retaining WGA, WGF, and ETA measurements.

Script: `benchmark_axton1986_micro.py`

Summary

Metric	This work	Report
Chi-squared	60.5	48
N measurements	112	101

Metric	This work	Report
N free parameters	37	38
Degrees of freedom	75	63

Fitted Parameters

Parameter	Fitted	Report	Diff%	Unc%	Rep Unc%	Match
SCA 33	12.3166	12.3204	-0.0308%	5.681	5.681	OK
SCA 35	15.8847	16.2388	-2.1804%	7.270	7.386	DIFF
SCA 39	7.9230	7.8966	+0.3348%	12.364	12.380	OK
SCA 41	11.9618	11.9740	-0.1017%	21.596	21.595	OK
SCR 33	11.1419	11.1480	-0.0543%	8.171	8.147	OK
SCR 35	14.0797	14.5259	-3.0719%	9.527	9.448	DIFF
SCR 39	6.9469	6.8131	+1.9637%	23.295	25.630	OK
SCR 41	10.8049	10.8285	-0.2178%	35.707	35.646	OK
ABS 33	575.2692	575.2391	+0.0052%	0.321	0.321	OK
ABS 35	682.0852	681.1651	+0.1351%	0.233	0.249	OK
ABS 39	1018.2345	1018.8813	-0.0635%	0.340	0.362	OK
ABS 41	1382.7102	1382.1872	+0.0378%	1.079	1.079	OK
FIS 33	530.3843	533.2694	-0.5410%	0.355	0.457	DIFF
FIS 35	584.0124	585.0603	-0.1791%	0.244	0.277	OK
FIS 39	748.1165	748.5181	-0.0536%	0.321	0.340	OK
FIS 41	1018.4694	1020.5836	-0.2072%	1.119	1.126	OK
CA 33	44.8849	41.9697	+6.9460%	1.837	4.182	DIFF
CA 35	98.0728	96.1049	+2.0477%	1.152	1.811	DIFF
CA 39	270.1180	270.3632	-0.0907%	0.913	1.174	OK
CA 41	364.2408	361.6036	+0.7293%	1.663	1.711	OK
NUB 33	2.4910	2.4856	+0.2189%	0.184	0.218	OK
NUB 35	2.4294	2.4261	+0.1348%	0.172	0.188	OK
NUB 39	2.8814	2.8794	+0.0683%	0.192	0.207	OK
NUB 41	2.9435	2.9406	+0.0989%	0.209	0.220	OK
NUB 52	3.7664	3.7644	+0.0518%	0.129	0.132	OK
ETA 33	2.2967	2.3042	-0.3264%	0.199	0.271	OK
ETA 35	2.0801	2.0838	-0.1792%	0.211	0.255	OK
ETA 39	2.1170	2.1153	+0.0802%	0.267	0.311	OK
ETA 41	2.1681	2.1713	-0.1468%	0.391	0.399	OK
ALPHA 33	0.0846	0.0787	+7.5313%	1.960	4.494	DIFF
ALPHA 35	0.1679	0.1643	+2.2089%	1.235	1.962	DIFF
ALPHA 39	0.3611	0.3612	-0.0376%	0.965	1.280	OK
ALPHA 41	0.3576	0.3543	+0.9414%	1.585	1.668	OK
WGA 33	0.9988	0.9988	+0.0040%	0.120	0.120	OK
WGA 35	0.9787	0.9787	+0.0000%	0.092	0.092	OK
WGA 39	1.0767	1.0767	+0.0000%	0.269	0.269	OK
WGA 41	1.0444	1.0444	+0.0000%	0.191	0.191	OK
WGF 33	0.9965	0.9966	-0.0140%	0.201	0.201	OK
WGF 35	0.9778	0.9778	+0.0000%	0.092	0.092	OK

Parameter	Fitted	Report	Diff%	Unc%	Rep Unc%	Match
WGF 39	1.0562	1.0562	+0.0000%	0.275	0.275	OK
WGF 41	1.0452	1.0452	+0.0000%	0.670	0.670	OK
F3ETA 33	741.9298	746.4586	-0.6067%	0.603	0.683	DIFF
F3ETA 35	719.7289	721.2735	-0.2141%	0.470	0.507	OK
F3ETA 39	1180.4095	1179.4821	+0.0786%	0.781	0.805	OK
F3ETA 41	1689.2737	1693.1349	-0.2280%	1.769	1.774	OK
CA 40	289.5000	289.3312	+0.0583%	0.484	0.482	OK
CA 42	18.5000	18.5148	-0.0799%	2.162	2.161	OK

Result: 39/47 parameters match. The chi-squared (60.5) and measurement count (112) both exceed the report's values (48 and 101), indicating that this narrow definition of "microscopic" includes 11 measurements not in the report's set. The most significant discrepancy is in FIS 33 (-0.54%) and the derived ALPHA 33 (+7.5%), suggesting that specific measurements biasing FIS 33 are included here but were excluded by Axton.

---

3. Microscopic Fit: Wide (Strict) Definition

To bracket the measurement set from the other direction, this benchmark uses a wider definition of "Maxwellian" that also excludes WGA, WGF, GC116, and GA116 measurements themselves (not just functions derived from them). This removes more measurements than necessary but tests the hypothesis that the discrepancy lies in the classification of WGA/WGF-type data.

Script: `benchmark_axton1986_micro_strict.py`

Summary

Metric	This work	Report
Chi-squared	60.5	48
N measurements	96	101
N free parameters	21	38
Degrees of freedom	75	63

Fitted Parameters

Parameter	Fitted	Report	Diff%	Unc%	Rep Unc%	Match
SCA 33	12.3166	12.3204	-0.0308%	5.681	5.681	OK
SCA 35	15.8847	16.2388	-2.1804%	7.270	7.386	DIFF
SCA 39	7.9230	7.8966	+0.3348%	12.364	12.380	OK
SCA 41	11.9618	11.9740	-0.1017%	21.596	21.595	OK
SCR 33	11.1419	11.1480	-0.0543%	8.171	8.147	OK
SCR 35	14.0797	14.5259	-3.0719%	9.527	9.448	DIFF
SCR 39	6.9469	6.8131	+1.9637%	23.295	25.630	OK
SCR 41	10.8049	10.8285	-0.2178%	35.707	35.646	OK

Parameter	Fitted	Report	Diff%	Unc%	Rep Unc%	Match
ABS 33	575.2692	575.2391	+0.0052%	0.321	0.321	OK
ABS 35	682.0852	681.1651	+0.1351%	0.233	0.249	OK
ABS 39	1018.2345	1018.8813	-0.0635%	0.340	0.362	OK
ABS 41	1382.7102	1382.1872	+0.0378%	1.079	1.079	OK
FIS 33	530.3843	533.2694	-0.5410%	0.355	0.457	DIFF
FIS 35	584.0124	585.0603	-0.1791%	0.244	0.277	OK
FIS 39	748.1165	748.5181	-0.0536%	0.321	0.340	OK
FIS 41	1018.4694	1020.5836	-0.2072%	1.119	1.126	OK
CA 33	44.8849	41.9697	+6.9460%	1.837	4.182	DIFF
CA 35	98.0728	96.1049	+2.0477%	1.152	1.811	DIFF
CA 39	270.1180	270.3632	-0.0907%	0.913	1.174	OK
CA 41	364.2408	361.6036	+0.7293%	1.663	1.711	OK
NUB 33	2.4910	2.4856	+0.2189%	0.184	0.218	OK
NUB 35	2.4294	2.4261	+0.1348%	0.172	0.188	OK
NUB 39	2.8814	2.8794	+0.0683%	0.192	0.207	OK
NUB 41	2.9435	2.9406	+0.0989%	0.209	0.220	OK
NUB 52	3.7664	3.7644	+0.0518%	0.129	0.132	OK
ETA 33	2.2967	2.3042	-0.3264%	0.199	0.271	OK
ETA 35	2.0801	2.0838	-0.1792%	0.211	0.255	OK
ETA 39	2.1170	2.1153	+0.0802%	0.267	0.311	OK
ETA 41	2.1681	2.1713	-0.1468%	0.391	0.399	OK
ALPHA 33	0.0846	0.0787	+7.5313%	1.960	4.494	DIFF
ALPHA 35	0.1679	0.1643	+2.2089%	1.235	1.962	DIFF
ALPHA 39	0.3611	0.3612	-0.0376%	0.965	1.280	OK
ALPHA 41	0.3576	0.3543	+0.9414%	1.585	1.668	OK

Result: 26/33 parameters match. Notably, this benchmark produces *identical* fitted values and chi-squared (60.5) to the narrow microscopic benchmark above, despite using only 96 measurements instead of 112. This is because WGA and WGF measurements (Nos. 54-61) are single-measurement constraints on their respective parameters — removing them simply leaves those parameters unconstrained without affecting the rest of the fit. The persistent FIS 33 discrepancy points to a different cause.

---

4. Microscopic Fit: Lounsbury Exclusion (Best Table 8 Match)

The key insight that resolves the Table 8 discrepancy comes from analysing the Lounsbury alpha measurements (Nos. 21-23). These measurements of (CA n)/FIS n for n = 33, 35, 39 have a systematic uncertainty code `Z_` ("Lounsbury Temperature Uncertainty") that involves reactor spectrum corrections, suggesting they may have been classified as integral/Maxwellian rather than microscopic despite measuring a ratio of microscopic quantities.

Lounsbury's alpha(233U) = 0.0861 is 2.3% higher than the converged value (0.0787), biasing FIS 33 downward when included. Excluding measurements 21-23 while including WGA/WGF (54-61) and CAP 34 (No. 5), and removing parameters not constrained without Maxwellian data (CA 40, CA 42, GC116, GA116), yields an almost perfect match with Table 8.

Script: `benchmark_axton1986_micro_no_lounsbury.py`

Summary

Metric	This work	Report
Chi-squared	55.1	48
N measurements	102	101
N free parameters	30	38
Degrees of freedom	72	63

Fitted Parameters

Parameter	Fitted	Report	Diff%	Unc%	Rep Unc%	Match
SCA 33	12.3206	12.3204	+0.0014%	5.681	5.681	OK
SCA 35	16.2235	16.2388	-0.0945%	7.385	7.386	OK
SCA 39	7.9001	7.8966	+0.0438%	12.380	12.380	OK
SCA 41	11.9720	11.9740	-0.0171%	21.596	21.595	OK
SCR 33	11.1483	11.1480	+0.0030%	8.161	8.147	OK
SCR 35	14.5068	14.5259	-0.1315%	9.455	9.448	OK
SCR 39	6.8449	6.8131	+0.4672%	24.890	25.630	OK
SCR 41	10.8234	10.8285	-0.0475%	35.520	35.646	OK
ABS 33	575.2413	575.2391	+0.0004%	0.321	0.321	OK
ABS 35	681.2060	681.1651	+0.0060%	0.249	0.249	OK
ABS 39	1018.7914	1018.8813	-0.0088%	0.362	0.362	OK
ABS 41	1382.2612	1382.1872	+0.0054%	1.079	1.079	OK
FIS 33	533.1364	533.2694	-0.0249%	0.460	0.457	OK
FIS 35	585.0334	585.0603	-0.0046%	0.277	0.277	OK
FIS 39	748.5245	748.5181	+0.0009%	0.340	0.340	OK
FIS 41	1020.3307	1020.5836	-0.0248%	1.127	1.126	OK
CA 33	42.1049	41.9697	+0.3221%	4.228	4.182	OK
CA 35	96.1726	96.1049	+0.0705%	1.811	1.811	OK
CA 39	270.2669	270.3632	-0.0356%	1.181	1.174	OK
CA 41	361.9304	361.6036	+0.0904%	1.729	1.711	OK
CAP 34	95.9000	95.8339	+0.0690%	2.086	2.071	OK
NUB 33	2.4860	2.4856	+0.0160%	0.217	0.218	OK
NUB 35	2.4264	2.4261	+0.0114%	0.188	0.188	OK
NUB 39	2.8794	2.8794	+0.0013%	0.207	0.207	OK
NUB 41	2.9406	2.9406	-0.0004%	0.220	0.220	OK
NUB 52	3.7649	3.7644	+0.0132%	0.132	0.132	OK
ETA 33	2.3040	2.3042	-0.0072%	0.269	0.271	OK
ETA 35	2.0838	2.0838	+0.0010%	0.255	0.255	OK
ETA 39	2.1156	2.1153	+0.0130%	0.310	0.311	OK
ETA 41	2.1706	2.1713	-0.0311%	0.399	0.399	OK
ALPHA 33	0.0790	0.0787	+0.3505%	4.545	4.494	OK
ALPHA 35	0.1644	0.1643	+0.0537%	1.962	1.962	OK
ALPHA 39	0.3611	0.3612	-0.0371%	1.289	1.280	OK

Parameter	Fitted	Report	Diff%	Unc%	Rep Unc%	Match
ALPHA 41	0.3547	0.3543	+0.1182%	1.697	1.668	OK
WGA 33	0.9988	0.9988	+0.0040%	0.120	0.120	OK
WGA 35	0.9787	0.9787	+0.0000%	0.092	0.092	OK
WGA 39	1.0767	1.0767	+0.0000%	0.269	0.269	OK
WGA 41	1.0444	1.0444	+0.0000%	0.191	0.191	OK
WGF 33	0.9965	0.9966	-0.0140%	0.201	0.201	OK
WGF 35	0.9778	0.9778	+0.0000%	0.092	0.092	OK
WGF 39	1.0562	1.0562	+0.0000%	0.275	0.275	OK
WGF 41	1.0452	1.0452	+0.0000%	0.670	0.670	OK
F3ETA 33	746.1097	746.4586	-0.0467%	0.680	0.683	OK
F3ETA 35	721.3019	721.2735	+0.0039%	0.507	0.507	OK
F3ETA 39	1179.5261	1179.4821	+0.0037%	0.803	0.805	OK
F3ETA 41	1692.3543	1693.1349	-0.0461%	1.773	1.774	OK

Result: 46/46 parameters match, including FIS 33 (533.14 vs 533.27, 0.025% difference) and ALPHA 33 (0.0790 vs 0.0787, 0.35%). This confirms that the Lounsbury measurements were treated as Maxwellian/integral in the original evaluation, likely because their systematic uncertainty (`Z_`: "Lounsbury Temperature Uncertainty") involves reactor spectrum corrections.

The remaining chi-squared gap (55.1 vs 48) and measurement count difference (102 vs 101) suggest one additional measurement was excluded by Axton. The residual differences are well within what can be attributed to PDF rounding of the input data.

---

5. Pipeline Validation: Modified JSON vs AGS (Exact Input Match)

Before comparing the two data pipelines on their actual (slightly different) data, we first verify that the JSON-based and AGS-based GLS pipelines produce identical results when given identical input. A modified JSON file (`axton1986_data_ags_mic.json`) was constructed by reading values from `thermalcst_axton1986_jan2026.mic` and the uncertainty decomposition from `ags_axton1986_jan2026.mic`, using an empirically verified column-to-correlation-code mapping (zero mismatches across all measurements). The covariance matrices from both representations agree to a relative Frobenius norm of 1.3e-9.

Script: `benchmark_modjson_vs_ags_mic.py`

Summary

Metric	Mod JSON	AGS	Report
Chi-squared	59.6738	59.6738	48
N measurements	102	102	101
N free parameters	29	29	38
Degrees of freedom	73	73	63

Fitted Parameters

Parameter	Mod JSON	AGS	Report	MJ-A diff%	MJ unc%	A unc%	R unc%
SCA 33	12.3205	12.3205	12.3204	-0.000000%	5.679	5.679	5.681
SCA 35	15.8264	15.8264	16.2388	+0.000000%	7.302	7.302	7.386
SCA 39	8.0178	8.0178	7.8966	-0.000000%	12.483	12.483	12.380
SCA 41	11.9450	11.9450	11.9740	+0.000000%	21.612	21.612	21.595
SCR 33	11.1454	11.1454	11.1480	-0.000000%	8.169	8.169	8.147
SCR 35	13.9984	13.9984	14.5259	+0.000001%	9.639	9.639	9.448
SCR 39	3.1317	3.1317	6.8131	-0.000026%	214.437	214.437	25.630
SCR 41	14.9731	14.9731	10.8285	+0.000008%	197.426	197.426	35.646
ABS 33	575.2960	575.2960	575.2391	+0.000000%	0.321	0.321	0.321
ABS 35	682.1918	682.1918	681.1651	-0.000000%	0.235	0.235	0.249
ABS 39	1020.1152	1020.1152	1018.8813	+0.000000%	0.460	0.460	0.362
ABS 41	1381.0269	1381.0269	1382.1872	-0.000000%	1.576	1.576	1.079
FIS 33	530.3722	530.3722	533.2694	-0.000000%	0.355	0.355	0.457
FIS 35	584.1181	584.1181	585.0603	+0.000000%	0.246	0.246	0.277
FIS 39	749.0977	749.0977	748.5181	+0.000000%	0.379	0.379	0.340
FIS 41	1017.1242	1017.1242	1020.5836	-0.000000%	1.573	1.573	1.126
NUB 33	2.4911	2.4911	2.4856	-0.000000%	0.184	0.184	0.218
NUB 35	2.4294	2.4294	2.4261	-0.000000%	0.172	0.172	0.188
NUB 39	2.8819	2.8819	2.8794	-0.000000%	0.194	0.194	0.207
NUB 41	2.9437	2.9437	2.9406	+0.000000%	0.209	0.209	0.220
NUB 52	3.7665	3.7665	3.7644	-0.000000%	0.129	0.129	0.132
WGA 33	0.9988	0.9988	0.9988	+0.000000%	0.120	0.120	0.120
WGA 35	0.9787	0.9787	0.9787	+0.000000%	0.092	0.092	0.092
WGA 39	1.0770	1.0770	1.0767	+0.000000%	0.269	0.269	0.269
WGA 41	1.0440	1.0440	1.0444	+0.000000%	0.191	0.191	0.191
WGF 33	0.9966	0.9966	0.9966	+0.000000%	0.201	0.201	0.201
WGF 35	0.9778	0.9778	0.9778	+0.000000%	0.092	0.092	0.092
WGF 39	1.0560	1.0560	1.0562	+0.000000%	0.275	0.275	0.275
WGF 41	1.0450	1.0450	1.0452	+0.000000%	0.670	0.670	0.670
CA 33	44.9238	44.9238	41.9697	+0.000000%	1.839	1.839	4.182
CA 35	98.0736	98.0736	96.1049	-0.000000%	1.153	1.153	1.811
CA 39	271.0175	271.0175	270.3632	-0.000000%	1.109	1.109	1.174
CA 41	363.9027	363.9027	361.6036	-0.000000%	2.093	2.093	1.711
ETA 33	2.2966	2.2966	2.3042	-0.000000%	0.199	0.199	0.271
ETA 35	2.0802	2.0802	2.0838	+0.000000%	0.211	0.211	0.255
ETA 39	2.1162	2.1162	2.1153	+0.000000%	0.286	0.286	0.311
ETA 41	2.1680	2.1680	2.1713	-0.000000%	0.392	0.392	0.399
ALPHA 33	0.0847	0.0847	0.0787	+0.000000%	1.963	1.963	4.494
ALPHA 35	0.1679	0.1679	0.1643	-0.000001%	1.236	1.236	1.962
ALPHA 39	0.3618	0.3618	0.3612	-0.000000%	1.063	1.063	1.280
ALPHA 41	0.3578	0.3578	0.3543	+0.000000%	1.595	1.595	1.668
F3ETA 33	742.1004	742.1004	746.4586	-0.000000%	0.603	0.603	0.683
F3ETA 35	719.9103	719.9103	721.2735	+0.000000%	0.472	0.472	0.507
F3ETA 39	1181.0189	1181.0188	1179.4821	+0.000000%	0.801	0.801	0.805

Parameter	Mod JSON	AGS	Report	MJ-A diff%	MJ unc%	A unc%	R unc%
F3ETA 41	1687.0712	1687.0712	1693.1349	-0.000000%	2.075	2.075	1.774

Result: All parameters agree to better than 3e-5%, chi-squared differs by 2.4e-6. The two pipelines are numerically equivalent. The 2 parameters nominally outside the 1e-6% threshold (SCR 39, SCR 41) have >200% uncertainties, making their tiny absolute differences register only due to the small denominator.

Note that SCR 39 and SCR 41 converge to unphysical values (3.13 and 14.97 barns vs the report's 6.81 and 10.83) with enormous uncertainties. This is because without macroscopic data, the coherent scattering cross sections SCR 39 and SCR 41 are essentially unconstrained — the microscopic measurements constrain only the incoherent scattering (SCA) and the total absorption. The report's fit with 38 parameters (vs our 29) apparently retained these as constrained through additional data or prior information not available in the microscopic subset alone.

---

6. Pipeline Comparison: Original JSON vs AGS (Actual Data Differences)

Having confirmed that both pipelines are numerically equivalent on identical input (Section 5), we now compare them using their actual, slightly different data. The JSON database was transcribed from the Axton 1986 PDF; the AGS/thermalcst files come from the CEA Cadarache digitization. Both are restricted to the same 102 microscopic measurement IDs.

Known differences between the two data sources:

- **Values:** 107 of 152 measurements differ by up to 0.05% due to rounding in the PDF transcription.
- **No. 147 uncertainty:** The AGS file has a total relative uncertainty of 0.399% vs the JSON's 0.380% (5% relative difference in the uncorrelated component).
- **Correlation code mapping:** The AGS files have a systematic 2-column offset for codes beyond position 50, causing 22 measurement pairs to be correlated in JSON but uncorrelated in AGS (primarily Popovic vs NUB measurements via the `Delta` code).
- **No. 14 (White, K_):** The JSON has K_ = 2.14% (from PDF), the AGS has 0.94% in the corresponding column, reducing the correlation between No. 14 and No. 100 from r = 0.79 to r = 0.46.

Script: `benchmark_json_vs_ags_mic.py`

Summary

Metric	JSON	AGS	Report
Chi-squared	60.0	59.7	48
N measurements	102	102	101
N free parameters	29	29	38
Degrees of freedom	73	73	63

Fitted Parameters

Parameter	JSON	AGS	Report	J-A diff%	J unc%	A unc%	R unc%
SCA 33	12.3166	12.3205	12.3204	-0.0322%	5.681	5.679	5.681
SCA 35	15.8239	15.8264	16.2388	-0.0157%	7.303	7.302	7.386
SCA 39	8.0167	8.0178	7.8966	-0.0130%	12.487	12.483	12.380
SCA 41	11.9397	11.9450	11.9740	-0.0444%	21.619	21.612	21.595
SCR 33	11.1419	11.1454	11.1480	-0.0318%	8.171	8.169	8.147
SCR 35	13.9950	13.9984	14.5259	-0.0244%	9.642	9.639	9.448
SCR 39	2.5957	3.1317	6.8131	-17.1160%	257.793	214.437	25.630
SCR 41	14.7685	14.9731	10.8285	-1.3667%	200.136	197.426	35.646
ABS 33	575.2695	575.2960	575.2391	-0.0046%	0.321	0.321	0.321
ABS 35	682.2074	682.1918	681.1651	+0.0023%	0.235	0.235	0.249
ABS 39	1020.3040	1020.1152	1018.8813	+0.0185%	0.457	0.460	0.362
ABS 41	1380.8315	1381.0269	1382.1872	-0.0141%	1.576	1.576	1.079
FIS 33	530.3586	530.3722	533.2694	-0.0026%	0.355	0.355	0.457
FIS 35	584.1272	584.1181	585.0603	+0.0016%	0.246	0.246	0.277
FIS 39	749.1111	749.0977	748.5181	+0.0018%	0.379	0.379	0.340
FIS 41	1016.9934	1017.1242	1020.5836	-0.0129%	1.574	1.573	1.126
NUB 33	2.4911	2.4911	2.4856	+0.0023%	0.184	0.184	0.218
NUB 35	2.4294	2.4294	2.4261	+0.0007%	0.172	0.172	0.188
NUB 39	2.8818	2.8819	2.8794	-0.0003%	0.194	0.194	0.207
NUB 41	2.9437	2.9437	2.9406	-0.0001%	0.209	0.209	0.220
NUB 52	3.7664	3.7665	3.7644	-0.0012%	0.129	0.129	0.132
WGA 33	0.9988	0.9988	0.9988	+0.0040%	0.120	0.120	0.120
WGA 35	0.9787	0.9787	0.9787	+0.0000%	0.092	0.092	0.092
WGA 39	1.0767	1.0770	1.0767	-0.0279%	0.269	0.269	0.269
WGA 41	1.0444	1.0440	1.0444	+0.0383%	0.191	0.191	0.191
WGF 33	0.9965	0.9966	0.9966	-0.0140%	0.201	0.201	0.201
WGF 35	0.9778	0.9778	0.9778	+0.0000%	0.092	0.092	0.092
WGF 39	1.0562	1.0560	1.0562	+0.0189%	0.275	0.275	0.275
WGF 41	1.0452	1.0450	1.0452	+0.0191%	0.670	0.670	0.670
CA 33	44.9109	44.9238	41.9697	-0.0285%	1.839	1.839	4.182
CA 35	98.0802	98.0736	96.1049	+0.0067%	1.152	1.153	1.811
CA 39	271.1929	271.0175	270.3632	+0.0647%	1.087	1.109	1.174
CA 41	363.8382	363.9027	361.6036	-0.0177%	2.093	2.093	1.711
ETA 33	2.2967	2.2966	2.3042	+0.0043%	0.199	0.199	0.271
ETA 35	2.0802	2.0802	2.0838	-0.0000%	0.211	0.211	0.255
ETA 39	2.1159	2.1162	2.1153	-0.0170%	0.278	0.286	0.311
ETA 41	2.1681	2.1680	2.1713	+0.0012%	0.392	0.392	0.399
ALPHA 33	0.0847	0.0847	0.0787	-0.0260%	1.963	1.963	4.494
ALPHA 35	0.1679	0.1679	0.1643	+0.0051%	1.236	1.236	1.962
ALPHA 39	0.3620	0.3618	0.3612	+0.0629%	1.040	1.063	1.280
ALPHA 41	0.3578	0.3578	0.3543	-0.0049%	1.595	1.595	1.668
F3ETA 33	741.9155	742.1004	746.4586	-0.0249%	0.603	0.603	0.683
F3ETA 35	719.9266	719.9103	721.2735	+0.0023%	0.472	0.472	0.507
F3ETA 39	1181.5885	1181.0188	1179.4821	+0.0482%	0.797	0.801	0.805

Parameter	JSON	AGS	Report	J-A diff%	J unc%	A unc%	R unc%
F3ETA 41	1686.9162	1687.0712	1693.1349	-0.0092%	2.075	2.075	1.774

Result: 41/45 well-constrained parameters agree within 0.05%. The 4 parameters outside this threshold are SCR 39, SCR 41 (unconstrained, >200% uncertainty), and WGA 39 and WGA 41 (0.03-0.04% difference, attributable to value rounding). The chi-squared difference (60.0 vs 59.7) reflects the cumulative effect of the small data differences.

The largest well-constrained difference is in CA 39 (0.065%), which propagates from the ABS 39 difference (0.019%) and the FIS 39 agreement (+0.002%). Overall, the two data sources produce negligibly different GLS results for all quantities of physical interest.

Conclusions

1. **All-data reproduction (Table 4):** Excellent agreement. Chi-squared 102.3 vs 103, with 47/54 parameters matching. The GLS implementation and data transcription are validated.
2. **Microscopic subset identification:** The report's Table 8 microscopic fit (101 measurements) excludes Lounsbury alpha measurements (Nos. 21-23), likely classified as integral due to reactor spectrum corrections. With this exclusion, all 46 reported parameters match within tolerance.
3. **Pipeline equivalence:** The JSON-based and AGS/thermalcst-based GLS pipelines produce identical results to machine precision (~1e-6%) when given identical input data, confirming both implementations are correct.
4. **Data source comparison:** The JSON (PDF-transcribed) and AGS (CEA Cadarache) data sources differ in value rounding (max 0.05%), one uncertainty (No. 147), and correlation code column mapping, but these differences have negligible impact on fitted parameters (<0.07% for all well-constrained quantities).

Update: Axton 1984 Paper Confirmation (April 2026)

The Axton 1984 paper (EUR 8805 EN), obtained through the NIST library on 2026-04-02, confirms the Lounsbury exclusion hypothesis.

Title: "Evaluation of the Thermal Neutron Constants of 233U, 235U, 239Pu and 241Pu and the Fission Neutron Yield of 252Cf," E.J. Axton, Central Bureau for Nuclear Measurements, JRC Geel.

Evidence from the paper:

1. **Table A1.2 (Function Definitions)** clearly distinguishes monoenergetic (2200 m/s) functions — ABS, FIS, SCA, SCR, NUB, HLF, WGA, WFA, GC116, GA116 — from Maxwellian (20C) derived functions — FA, FF, CAP, FFH, ETA, F1ETA, F2ETA, F3ETA, FLEM, F1BIG, F1CAB-F5CAB. This matches exactly the `MAXWELLIAN_DERIVED_FUNCS` set used in our benchmark scripts.

2. **Table 6 ("Maxwellian Data Omitted")** shows the microscopic-only fit results. The fitted ALPHA 33 = 0.0785, consistent with our no-Lounsbury result of 0.0787 and far below Beer's 0.0861. If Lounsbury had been included, this value would be pulled significantly upward toward 0.085.

3. **Table 3** confirms that measurements 21-23 (Lounsbury 1970) carry the uncertainty code **"Z"**, which Table 2 defines as **"LOUNSBURY TEMPERATURE"** — a systematic Maxwellian temperature correction.

4. **Appendix 4, Note 19** states: "Based on Monte Carlo evaluation (Beer et al (1975)). Correlated through temperature uncertainty." This confirms Axton used the Beer-corrected values and that the correlation arises from reactor spectrum temperature.

The paper thus validates our hypothesis: the Lounsbury measurements use monoenergetic function expressions (CA/FIS) but were measured in a reactor spectrum requiring Maxwellian temperature corrections. Axton classified them as effectively Maxwellian and excluded them from Table 6/Table 8.

Update: Lounsbury and Arif Sensitivity Studies (April 2026)

Motivation

During discussions among the TNC collaborators (March-April 2026):

- It was argued that Lounsbury/Beer data should be included because "without Lounsbury/Beer we cannot constrain U-233 capture." The Beer-corrected alpha(233U) = 0.0861 +/- 0.0021, the current standard alpha = 0.0842 is within 1-sigma, while our 0.079 without Lounsbury is clearly outside.
- GMA fit comparisons were provided showing sensitivity to both the Lounsbury/Beer data and the Arif U-235 scattering measurement, presented in a 2x2 matrix (with/without each).
- It was noted that the TNC database had been updated for NDS2025 and that older presentation numbers should not be used.

We ran a systematic sensitivity study using the Axton 1986 microscopic GLS framework.

Scenario B: Microscopic fit with Lounsbury included

Script: `benchmark_axton1986_micro_with_lounsbury.py`

Adding Lounsbury measurements (21-23) back to the microscopic baseline (105 measurements, 30 free parameters):

Parameter	Without Lounsbury (A)	With Lounsbury (B)	Shift	Beer value
FIS 33	533.14	530.38	-2.76 barn	
CA 33	42.10	44.88	+2.78 barn	
ABS 33	575.24	575.27	+0.03 barn	
ALPHA 33	0.0790	0.0846	+7.1%	0.0861
FIS 35	585.03	584.01	-1.02 barn	
CA 35	96.17	98.07	+1.90 barn	

Parameter	Without Lounsbury (A)	With Lounsbury (B)	Shift	Beer value
ALPHA 35	0.1644	0.1679	+2.1%	0.1697
FIS 39	748.52	748.12	-0.40 barn	
ALPHA 39	0.3611	0.3611	-0.0%	0.3555
Chi-sq.	55.1 (72 dof)	60.5 (75 dof)		

Mechanism: Absorption stays nearly constant (~575.2 barn for U-233) while ~2.8 barn is redistributed from fission to capture. Chi-squared per dof remains acceptable (0.81 vs 0.76), with no severe tension.

Lounsbury residuals:

No.	Quantity	Measured	Fitted	Residual
21	alpha(233U)	0.0861	0.0846	+1.7%
22	alpha(235U)	0.1697	0.1679	+1.0%
23	alpha(239Pu)	0.3555	0.3611	-1.6%

The argument for inclusion is validated: the fitted alpha(233U) = 0.085 agrees well with the current standard (0.084), while excluding Lounsbury gives 0.079 which is outside the 1-sigma range of Beer's value.

Arif SCA(35) measurement

The Arif measurement (M. Arif et al., Phys. Rev. A35, 2810 (1987)):

- **SCA(35) = 14.0 barn**, uncertainty 1.6%
- From neutron interferometry (coherent scattering length)
- Much more precise than Lemmel (1982) SCA(35) = 16.508 barn at 7.9%
- Not in the original Axton 1986 database (post-1986 measurement)
- Found in this project's `expdata.py` at commit `807aafb` (No. 173) and in the neutron standards evaluation database (NS=1918, channel SS-U5 = U-235 elastic scattering)

The Arif measurement was injected as an additional uncorrelated data point. Because Arif (1.6%) is ~5x more precise than Lemmel (7.9%), it dominates the SCA(35) fit completely.

Complete 2x2 matrix: Lounsbury x Arif

Script: `benchmark_axton1986_lounsbury_arif_matrix.py`

All four scenarios using the Axton 1986 microscopic GLS framework:

```
                  No Arif          With Arif
               +----------------+----------------+
 No Lounsbury  | A) 102 meas    | C) 103 meas    |
               | chi2 = 55.1    | chi2 = 59.9    |
               +----------------+----------------+
 With Lounsbury| B) 105 meas    | D) 106 meas    |
               | chi2 = 60.5    | chi2 = 64.3    |
               +----------------+----------------+
```

Key parameters across all four cells:

Parameter	(A) baseline	(B) +Loun.	(C) +Arif	(D) +both	GMA +both	GMA -both
FIS 33	533.14	530.38	532.83	530.36	532.70	533.90
FIS 35	585.03	584.01	585.80	584.91	587.28	586.50
FIS 39	748.52	748.12	748.76	748.48	750.30	749.70
CA 33	42.10	44.88	42.41	44.91	44.76	42.50
CA 35	96.17	98.07	97.20	98.55	99.50	98.90
CA 39	270.27	270.12	270.65	270.22	269.70	270.60
ABS 35	681.21	682.09	682.99	683.46	-	-
ALPHA 33	0.0790	0.0846	0.0796	0.0847	0.0840	0.0788
ALPHA 35	0.1644	0.1679	0.1659	0.1685	0.1694	0.1687
ALPHA 39	0.3611	0.3611	0.3615	0.3610	0.3595	0.3610
SCA 35	16.22	15.88	14.00	14.00	-	-

The GMA columns are results from a collaborator's GMA fit (discussed 2026-04-03/04) using the Standards 2017 database plus post-2017 measurements.

Marginal effects (deviations from baseline A):

Parameter	Lounsbury (B-A)	Arif (C-A)	Combined (D-A)	Interaction (D-B-C+A)
FIS 33	-2.75	-0.30	-2.78	+0.28
FIS 35	-1.02	+0.76	-0.12	+0.13
CA 33	+2.78	+0.31	+2.81	-0.28
CA 35	+1.90	+1.02	+2.38	-0.54
ABS 35	+0.88	+1.79	+2.26	-0.41
ALPHA 33	+0.0057	+0.0006	+0.0057	-0.0006
ALPHA 35	+0.0035	+0.0015	+0.0041	-0.0010
SCA 35	-0.34	-2.22	-2.22	+0.34

Conclusions from the sensitivity study

1. **Lounsbury dominates alpha(233U):** shift of +0.006 (0.079 to 0.085). Arif contributes only +0.0006, an order of magnitude smaller.
2. **Arif's main effect is on U-235:** ABS(35) increases by +1.8 barn through the lower scattering cross section, split between CA(35) (+1.0) and FIS(35) (+0.8). The Lounsbury effect on U-235 is comparable in magnitude but opposite in direction for FIS(35).
3. **The effects are largely additive.** Interaction terms are small (typically <15% of the main effect), meaning the Lounsbury and Arif effects can be discussed independently.
4. **Good agreement with the GMA fit.** Our "without both" (A) gives alpha(233U) = 0.0790 vs GMA 0.0788. Our "with both" (D) gives 0.0847 vs GMA 0.0840. The trends and magnitudes match well despite different databases (Axton 1986 vs Standards 2017 + post-2017 measurements).

5. **The argument for inclusion is supported.** Including Lounsbury brings alpha(233U) from 0.079 to 0.085, consistent with the current standard (0.084) and within the Beer uncertainty band. The fit absorbs the Lounsbury data without serious tension (chi2/dof = 0.81).

Updated benchmark inventory

Script	Description
`benchmark_axton1986_all_data.py`	Table 4 reproduction (167 meas)
`benchmark_axton1986_micro.py`	Narrow microscopic (112 meas)
`benchmark_axton1986_micro_strict.py`	Strict microscopic (96 meas)
`benchmark_axton1986_micro_no_lounsbury.py`	Best Table 8 match (102 meas)
`benchmark_axton1986_micro_with_lounsbury.py`	+ Lounsbury (105 meas)
`benchmark_axton1986_micro_with_lounsbury_arif.py`	+ Lounsbury + Arif (106 meas)
`benchmark_axton1986_lounsbury_arif_matrix.py`	2x2 matrix x Arif comparison